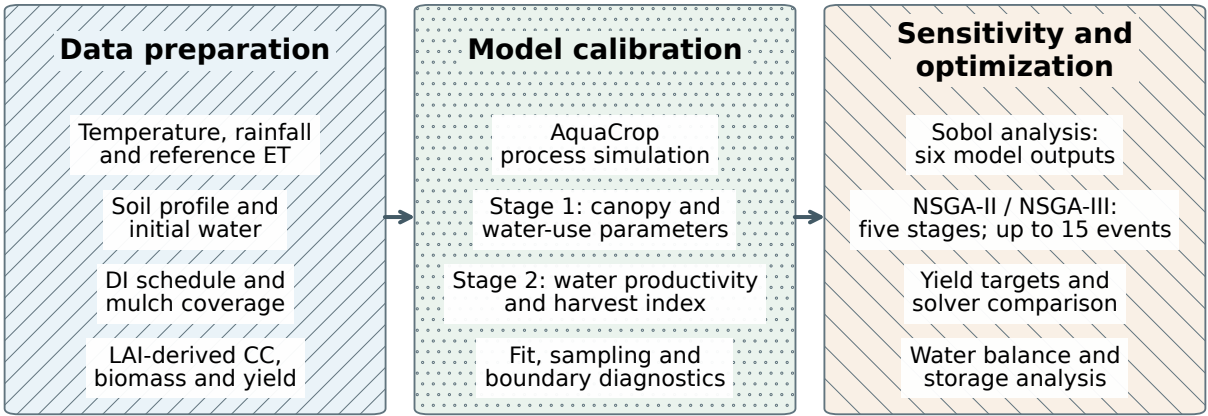
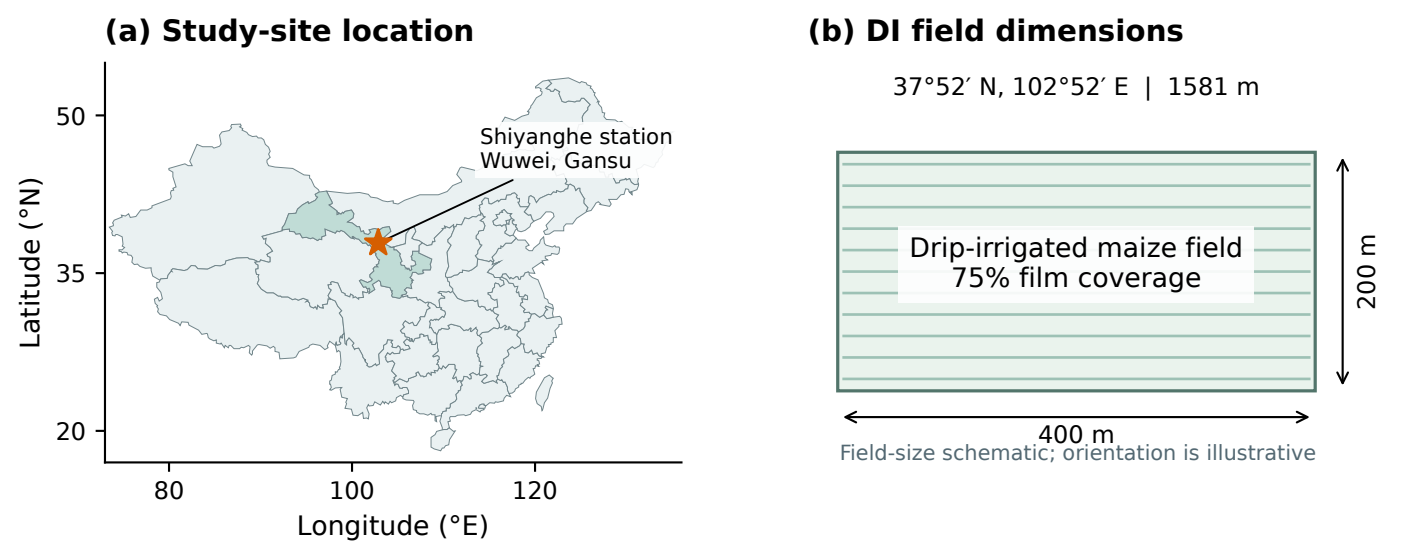
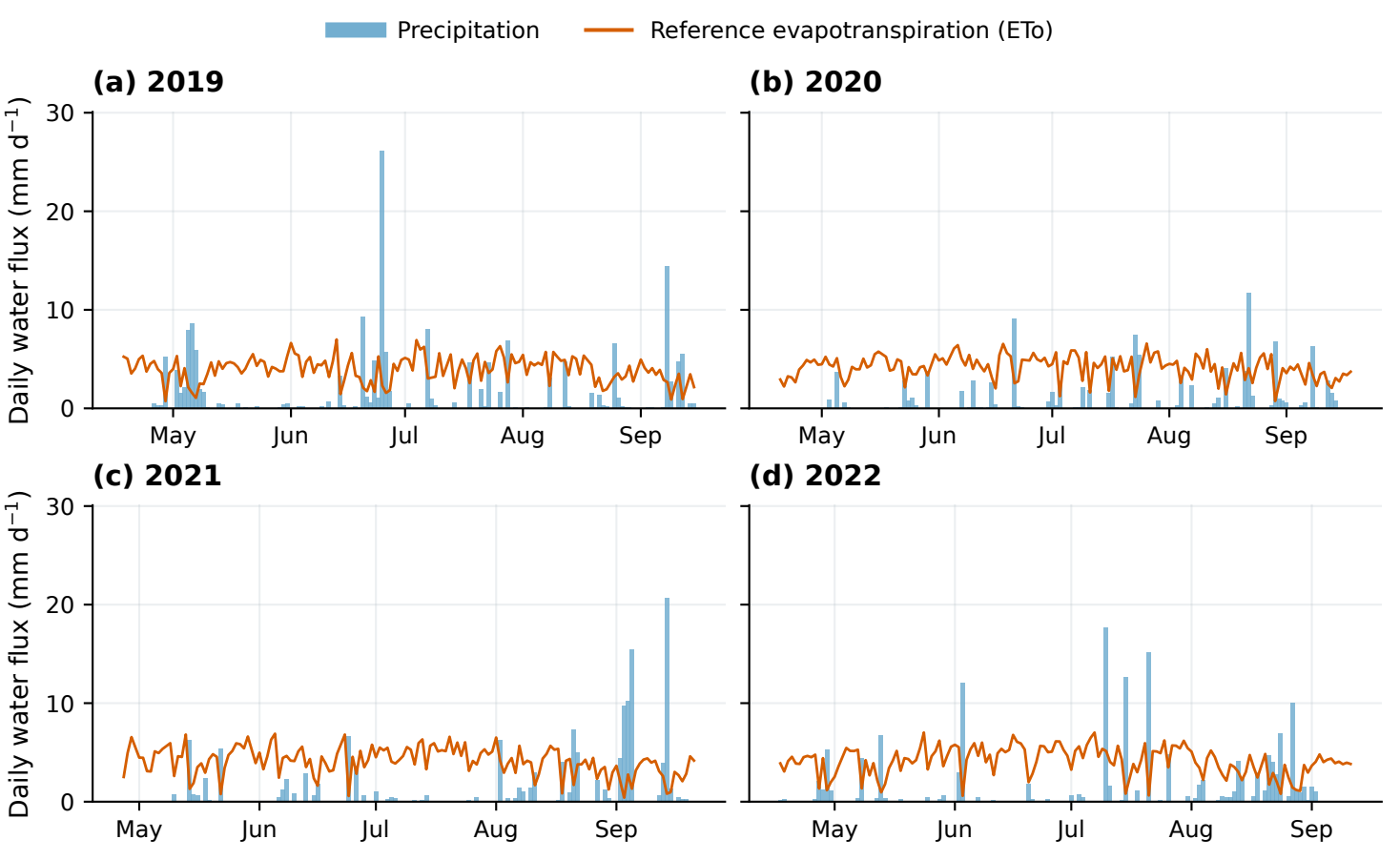
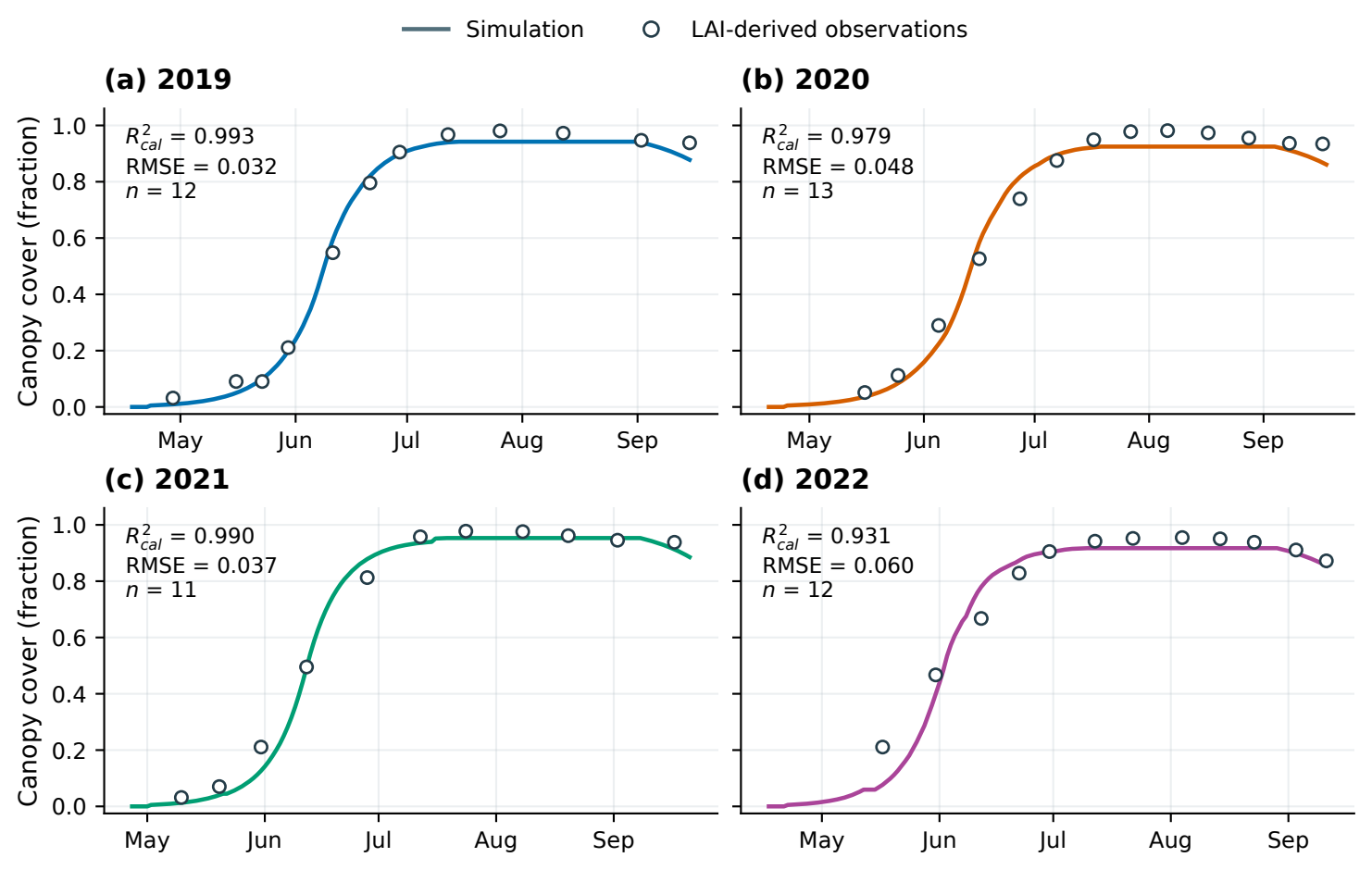
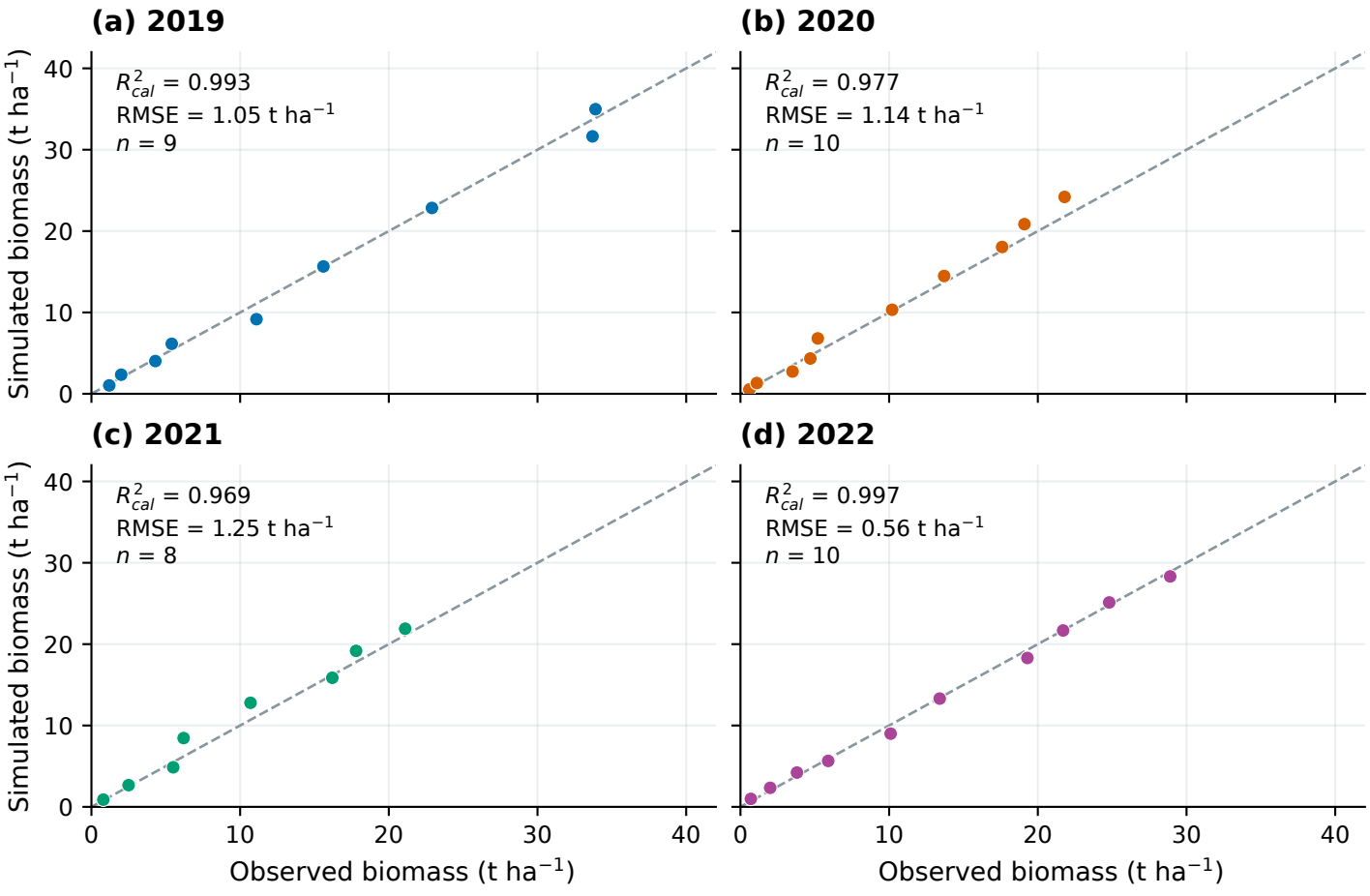
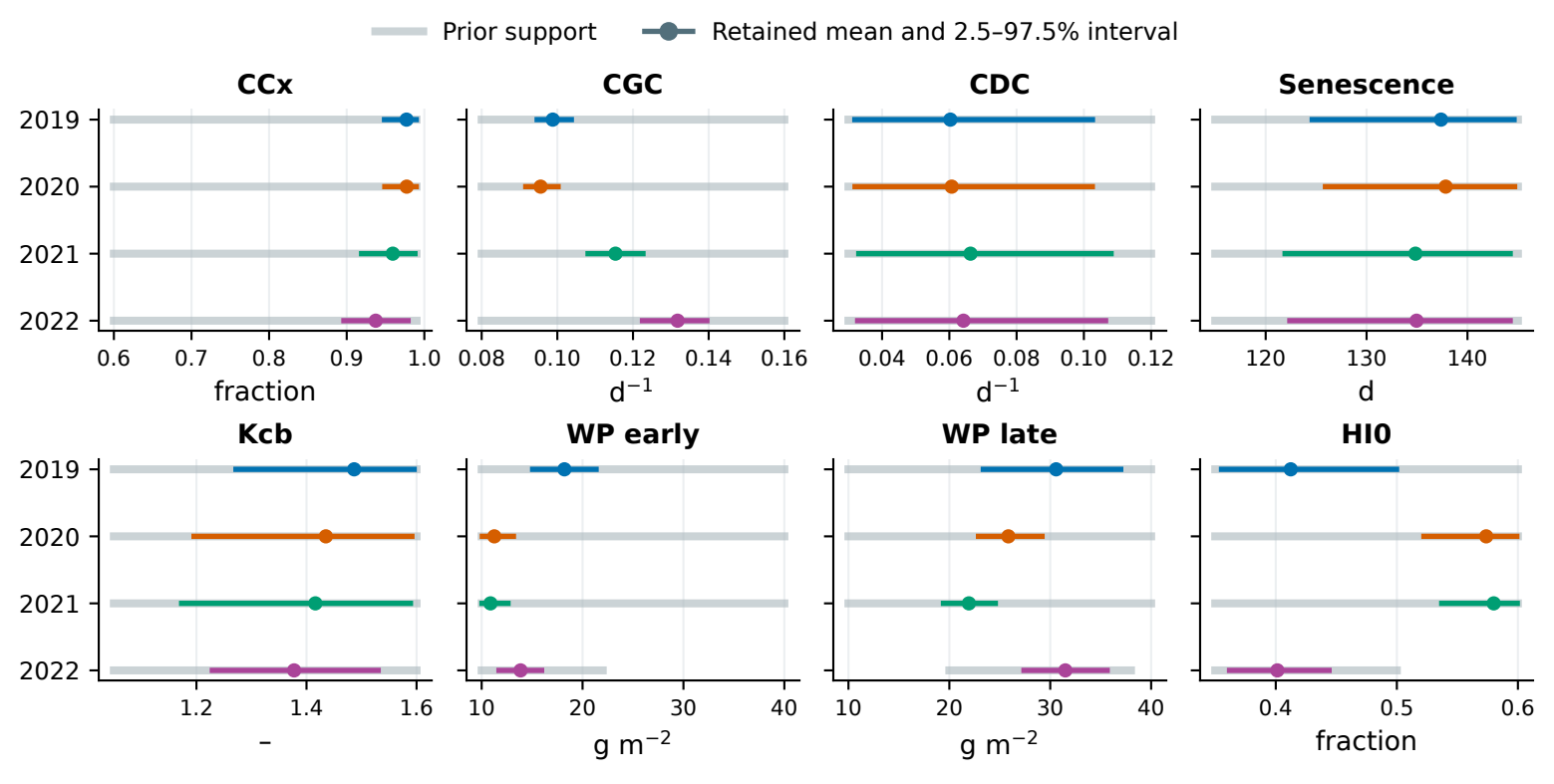

Data preparation
Temperature, rainfall and reference ET
Soil profile and initial water
DI schedule and mulch coverage
LAI-derived CC, biomass and yield
Model calibration
AquaCrop process simulation
Stage 1: canopy and water-use parameters
Stage 2: water productivity and harvest index
Fit, sampling and boundary diagnostics
Sensitivity and optimization
Sobol analysis: six model outputs
NSGA-II / NSGA-III: five stages; up to 15 events
Yield targets and solver comparison
Water balance and storage analysis

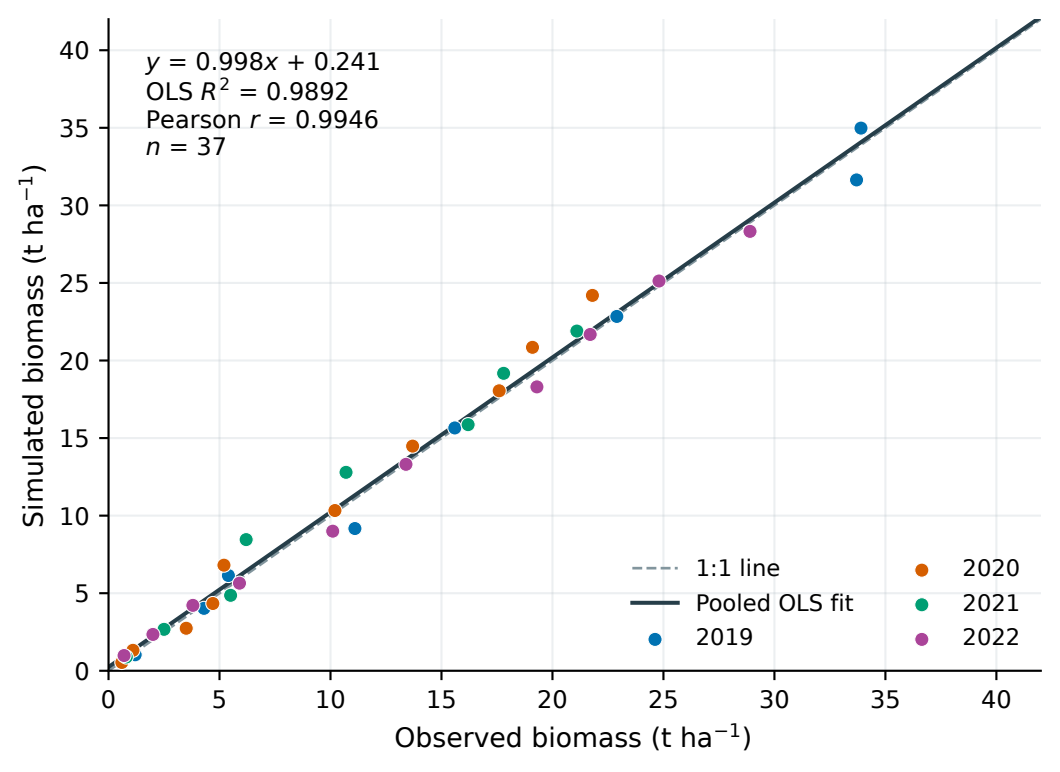

(a) Study-site location
Shiyanghe station
Wuwei, Gansu
Latitude (°N)
Longitude (°E)
50
35
20
80
100
120
(b) DI field dimensions
37°52′ N, 102°52′ E | 1581 m
Drip-irrigated maize field
75% film coverage
200 m
400 m
Field-size schematic; orientation is illustrative

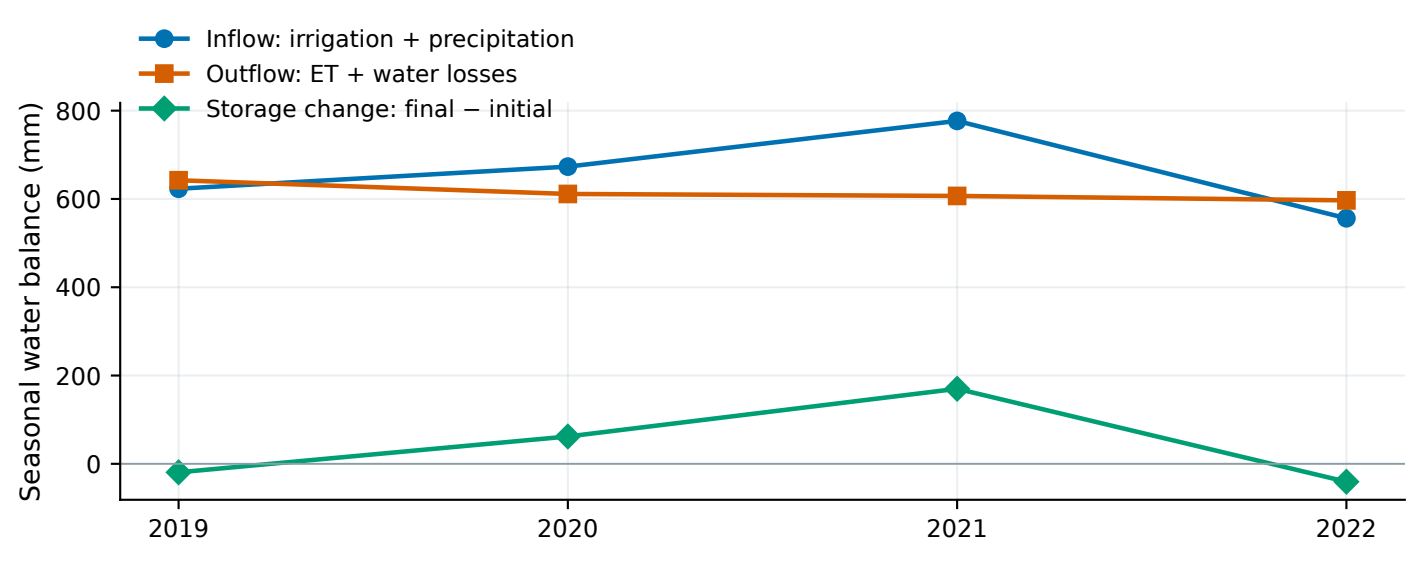

Precipitation
Reference evapotranspiration (ETo)
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Daily water flux (mm d^{-1})
0
10
20
30
May
Jun
Jul
Aug
Sep

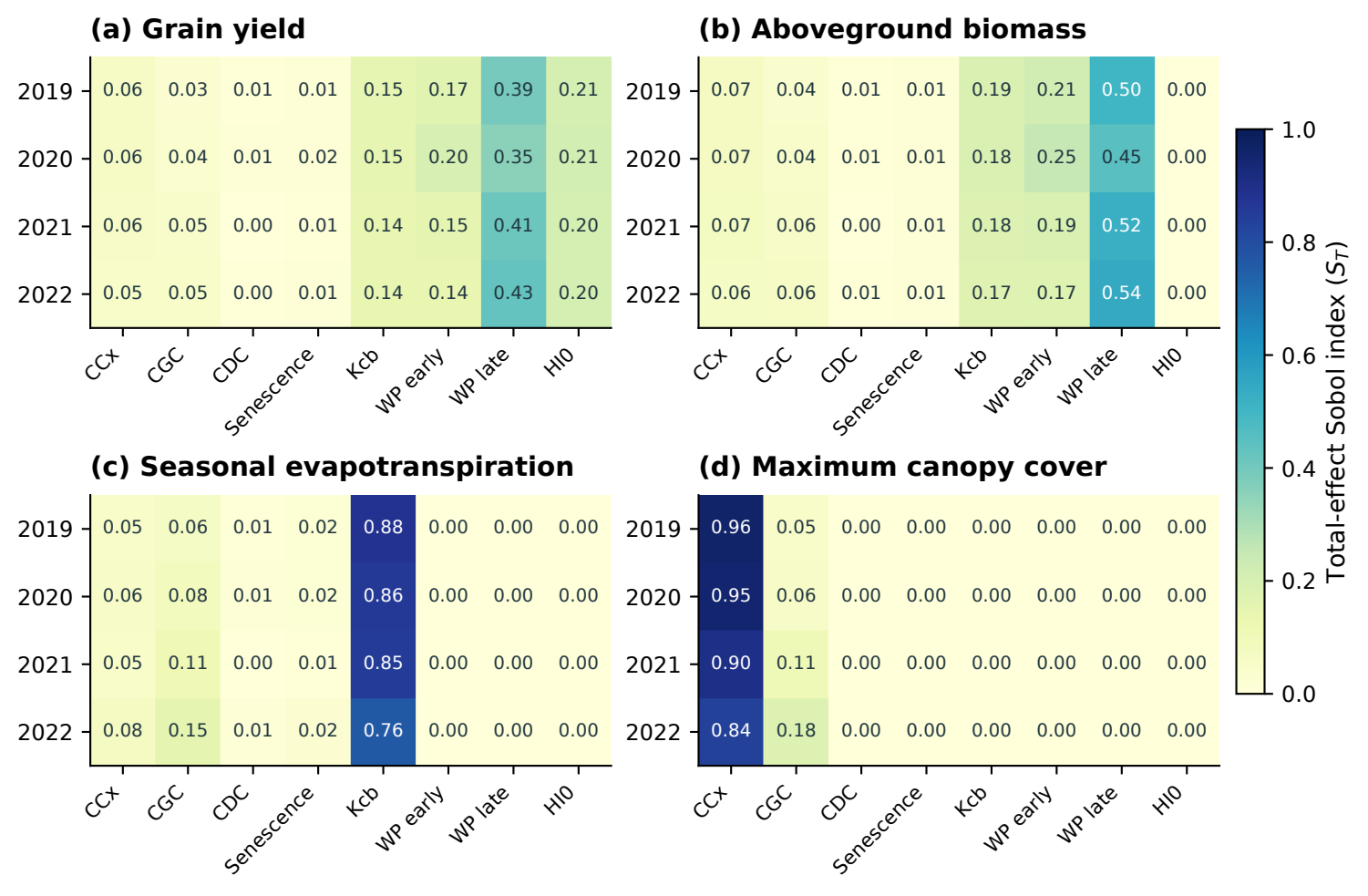

Simulation
LAI-derived observations
(a) 2019
R^2_{cal} = 0.993
RMSE = 0.032
n = 12
(b) 2020
R^2_{cal} = 0.979
RMSE = 0.048
n = 13
(c) 2021
R^2_{cal} = 0.990
RMSE = 0.037
n = 11
(d) 2022
R^2_{cal} = 0.931
RMSE = 0.060
n = 12
Canopy cover (fraction)
0.0
0.2
0.4
0.6
0.8
1.0
May
Jun
Jul
Aug
Sep

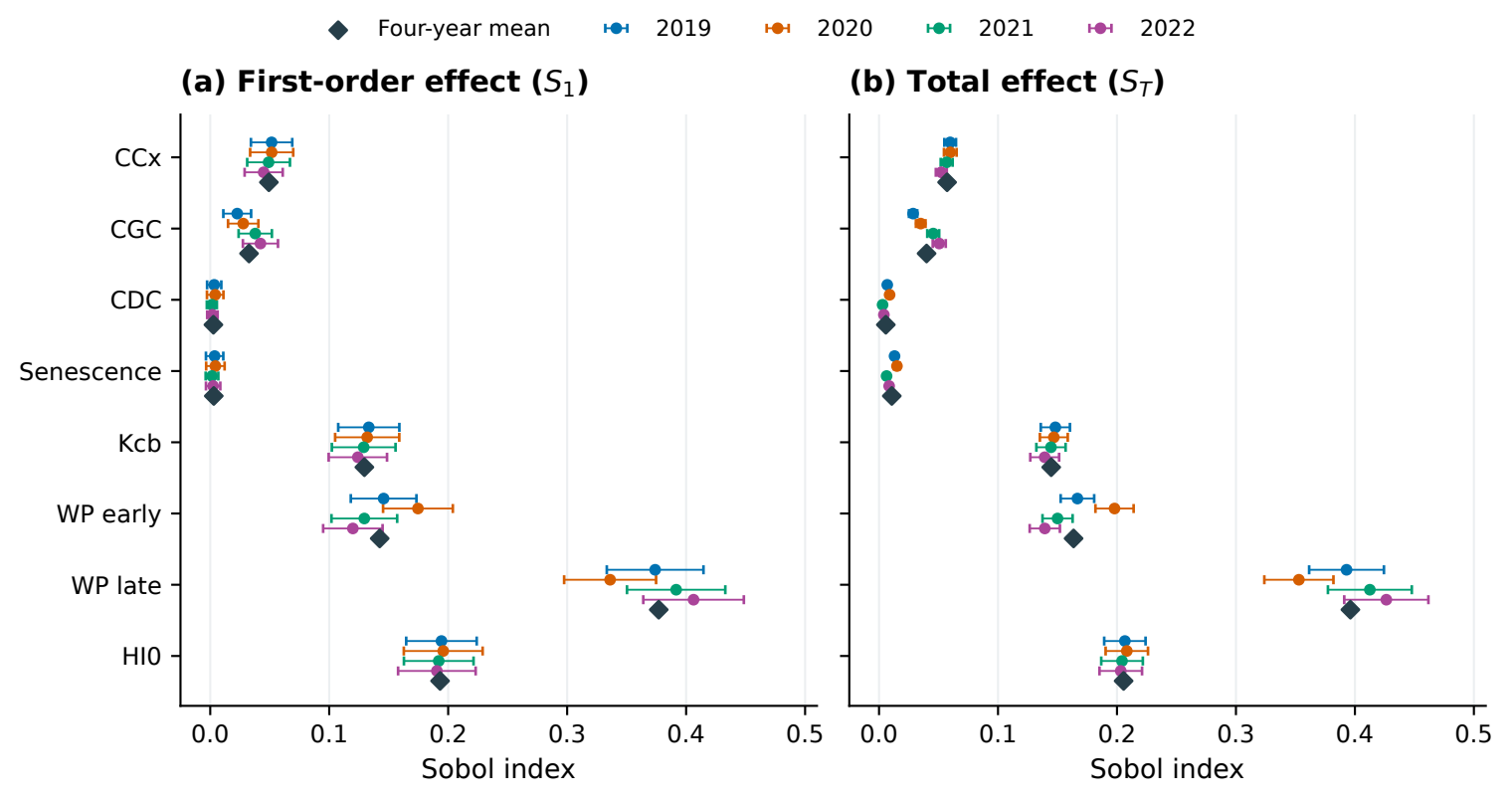

(a) 2019
R^2_{cal} = 0.993
RMSE = 1.05 t ha^{-1}
n = 9
(b) 2020
R^2_{cal} = 0.977
RMSE = 1.14 t ha^{-1}
n = 10
(c) 2021
R^2_{cal} = 0.969
RMSE = 1.25 t ha^{-1}
n = 8
(d) 2022
R^2_{cal} = 0.997
RMSE = 0.56 t ha^{-1}
n = 10
Simulated biomass (t ha^{-1})
Observed biomass (t ha^{-1})
0
10
20
30
40

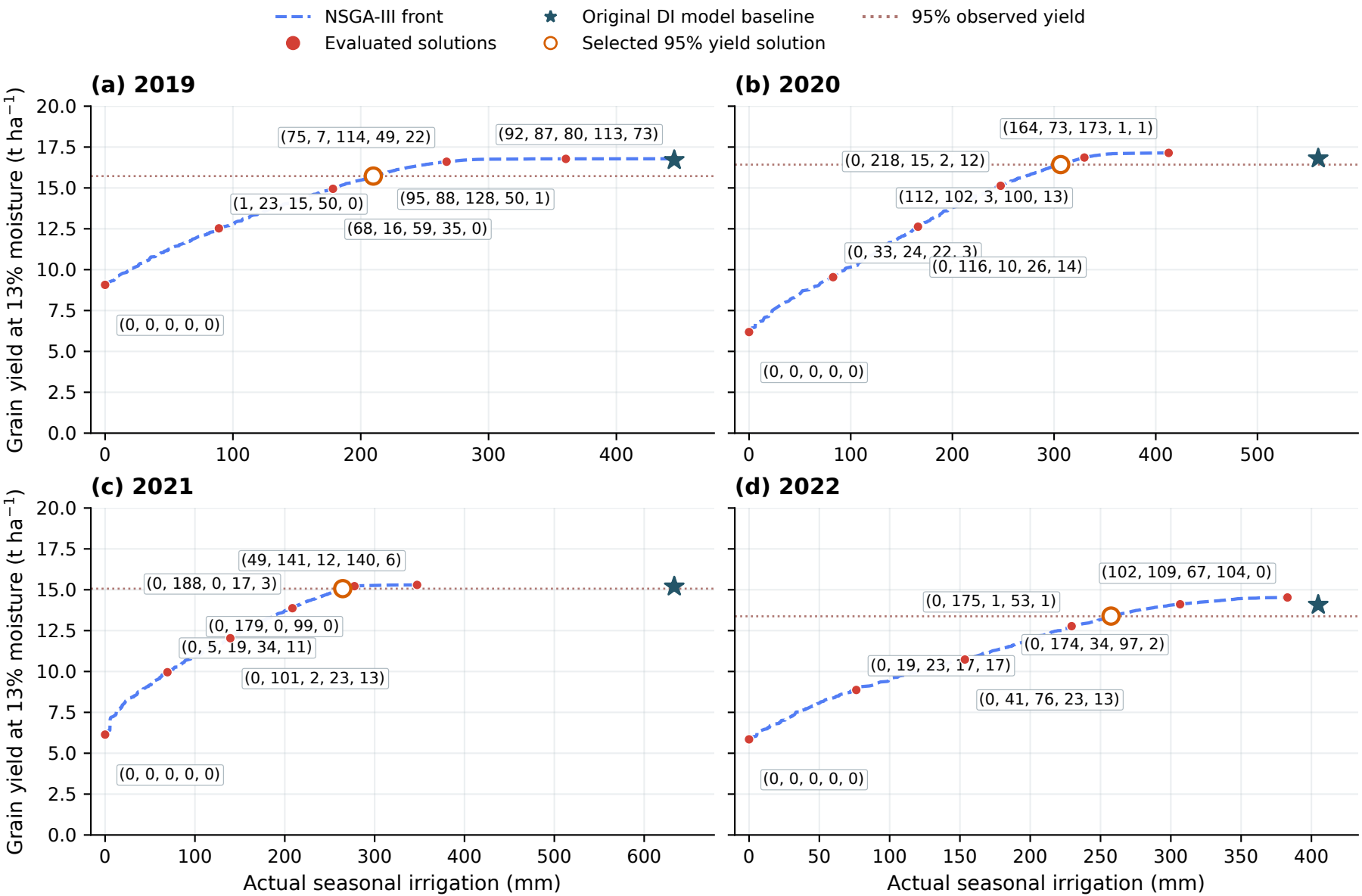

Prior support
Retained mean and 2.5–97.5% interval
CCx
CGC
CDC
Senescence
Kcb
WP early
WP late
HI0
2019
2020
2021
2022
0.6 0.7 0.8 0.9 1.0
fraction
0.08 0.10 0.12 0.14 0.16
d^{-1}
0.04 0.06 0.08 0.10 0.12
d^{-1}
120 130 140
d
1.2 1.4 1.6
–
10 20 30 40
g m^{-2}
10 20 30 40
g m^{-2}
0.4 0.5 0.6
fraction

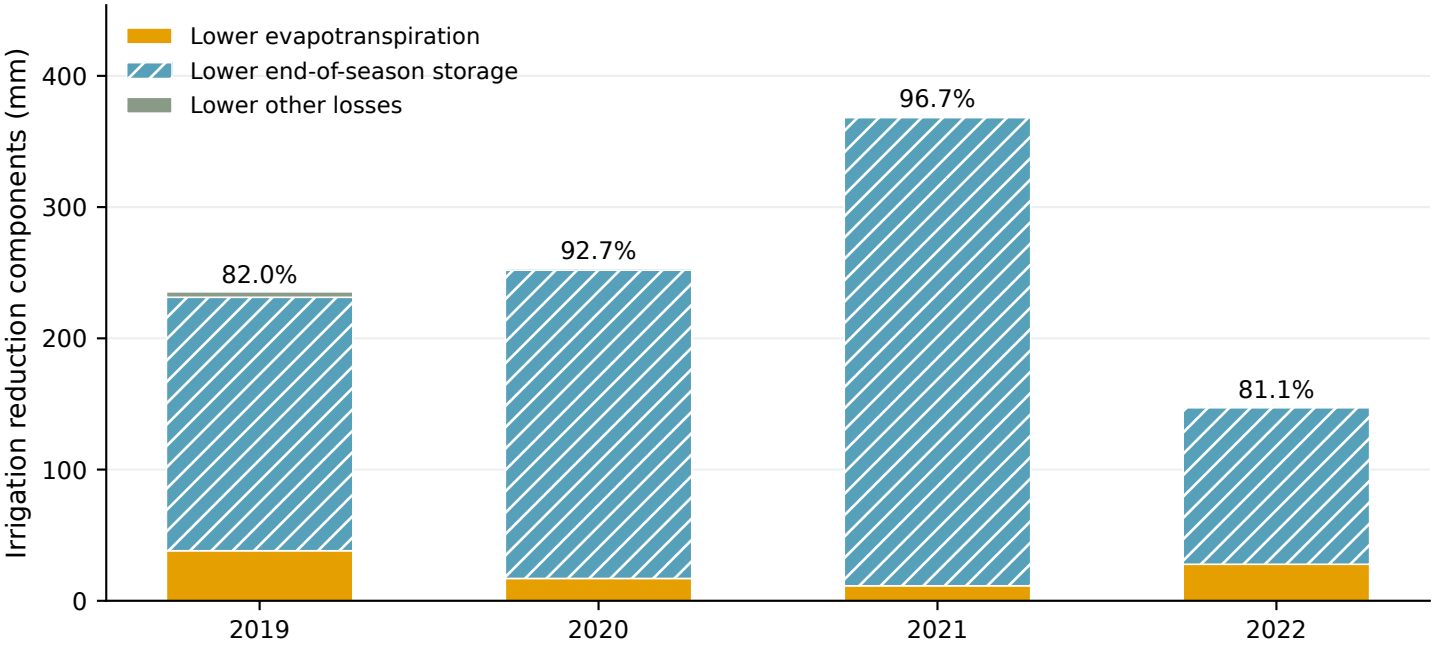

$y = 0.998x + 0.241$
OLS $R^2 = 0.9892$
Pearson $r = 0.9946$
$n = 37$
1:1 line
Pooled OLS fit
2019
2020
2021
2022
Simulated biomass (t ha^{-1})
Observed biomass (t ha^{-1})
0
5
10
15
20
25
30
35
40

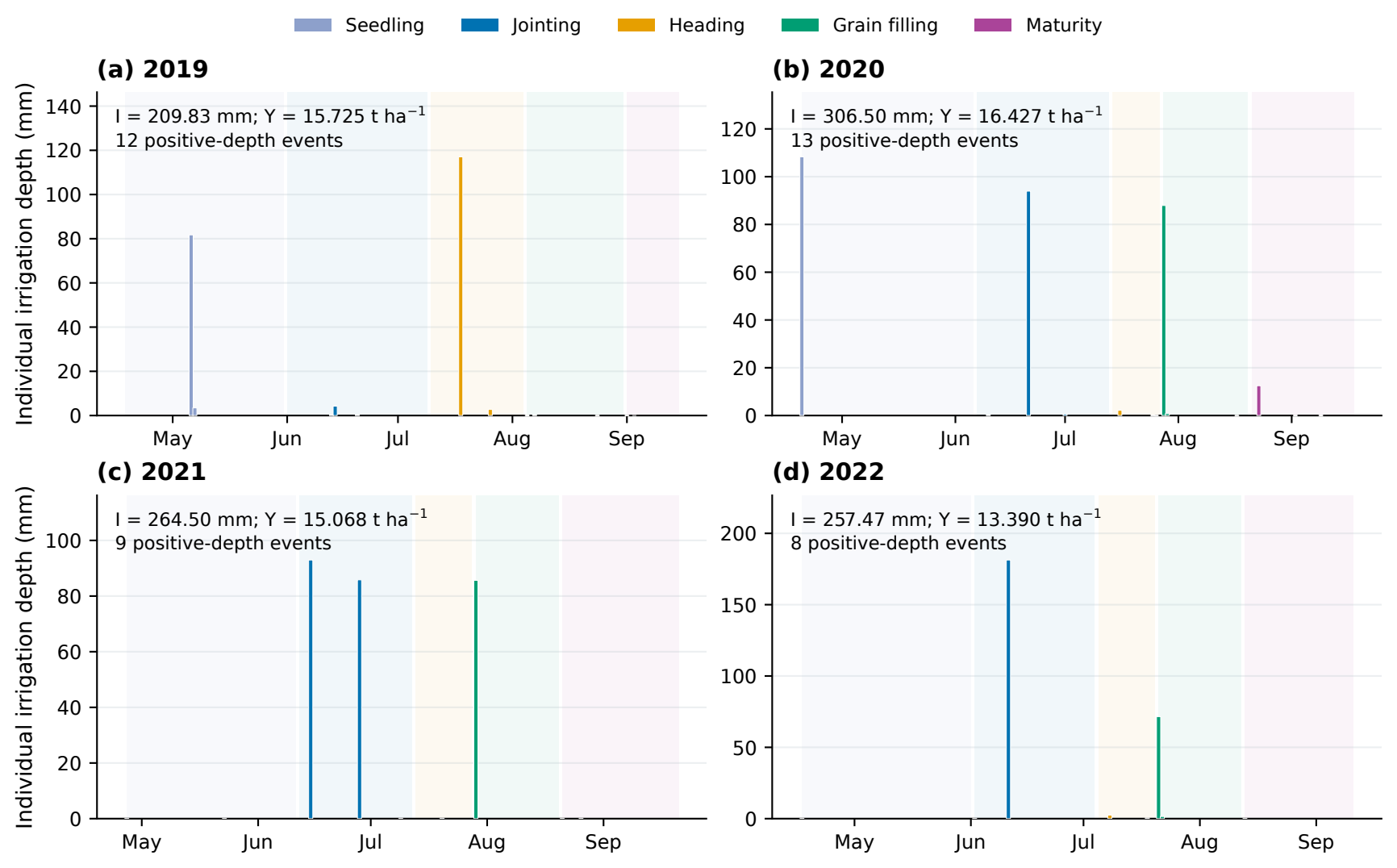

Inflow: irrigation + precipitation
Outflow: ET + water losses
Storage change: final − initial
Seasonal water balance (mm)
800
600
400
200
0
2019
2020
2021
2022

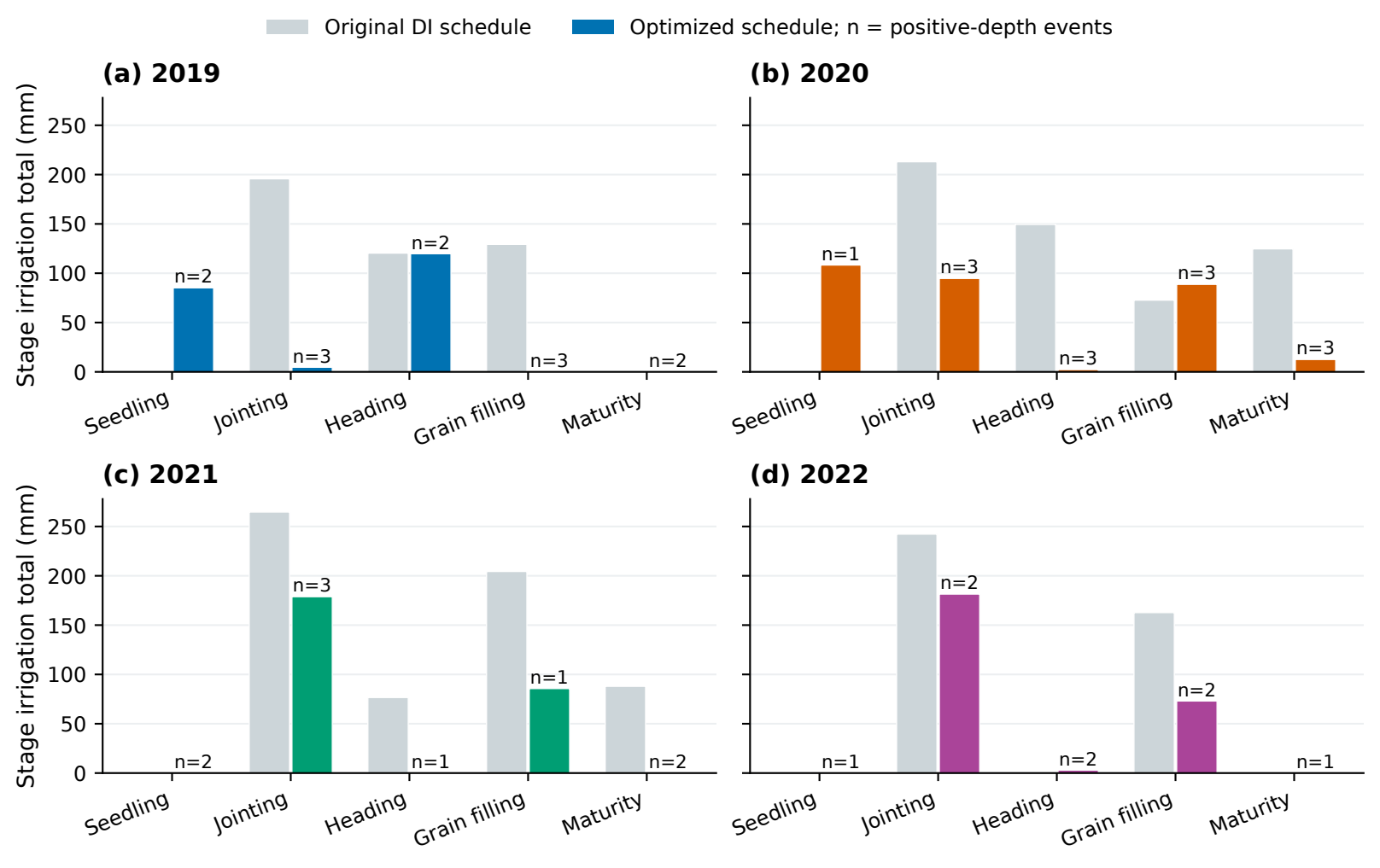

(a) Grain yield
2019 0.06 0.03 0.01 0.01 0.15 0.17 0.39 0.21
2020 0.06 0.04 0.01 0.02 0.15 0.20 0.35 0.21
2021 0.06 0.05 0.00 0.01 0.14 0.15 0.41 0.20
2022 0.05 0.05 0.00 0.01 0.14 0.14 0.43 0.20
(b) Aboveground biomass
2019 0.07 0.04 0.01 0.01 0.19 0.21 0.50 0.00
2020 0.07 0.04 0.01 0.01 0.18 0.25 0.45 0.00
2021 0.07 0.06 0.00 0.01 0.18 0.19 0.52 0.00
2022 0.06 0.06 0.01 0.01 0.17 0.17 0.54 0.00
(c) Seasonal evapotranspiration
2019 0.05 0.06 0.01 0.02 0.88 0.00 0.00 0.00
2020 0.06 0.08 0.01 0.02 0.86 0.00 0.00 0.00
2021 0.05 0.11 0.00 0.01 0.85 0.00 0.00 0.00
2022 0.08 0.15 0.01 0.02 0.76 0.00 0.00 0.00
(d) Maximum canopy cover
2019 0.96 0.05 0.00 0.00 0.00 0.00 0.00 0.00
2020 0.95 0.06 0.00 0.00 0.00 0.00 0.00 0.00
2021 0.90 0.11 0.00 0.00 0.00 0.00 0.00 0.00
2022 0.84 0.18 0.00 0.00 0.00 0.00 0.00 0.00
CCx CGC CDC Senescence Kcb WP early WP late HI0
Total-effect Sobol index (S_T)
1.0 0.8 0.6 0.4 0.2 0.0

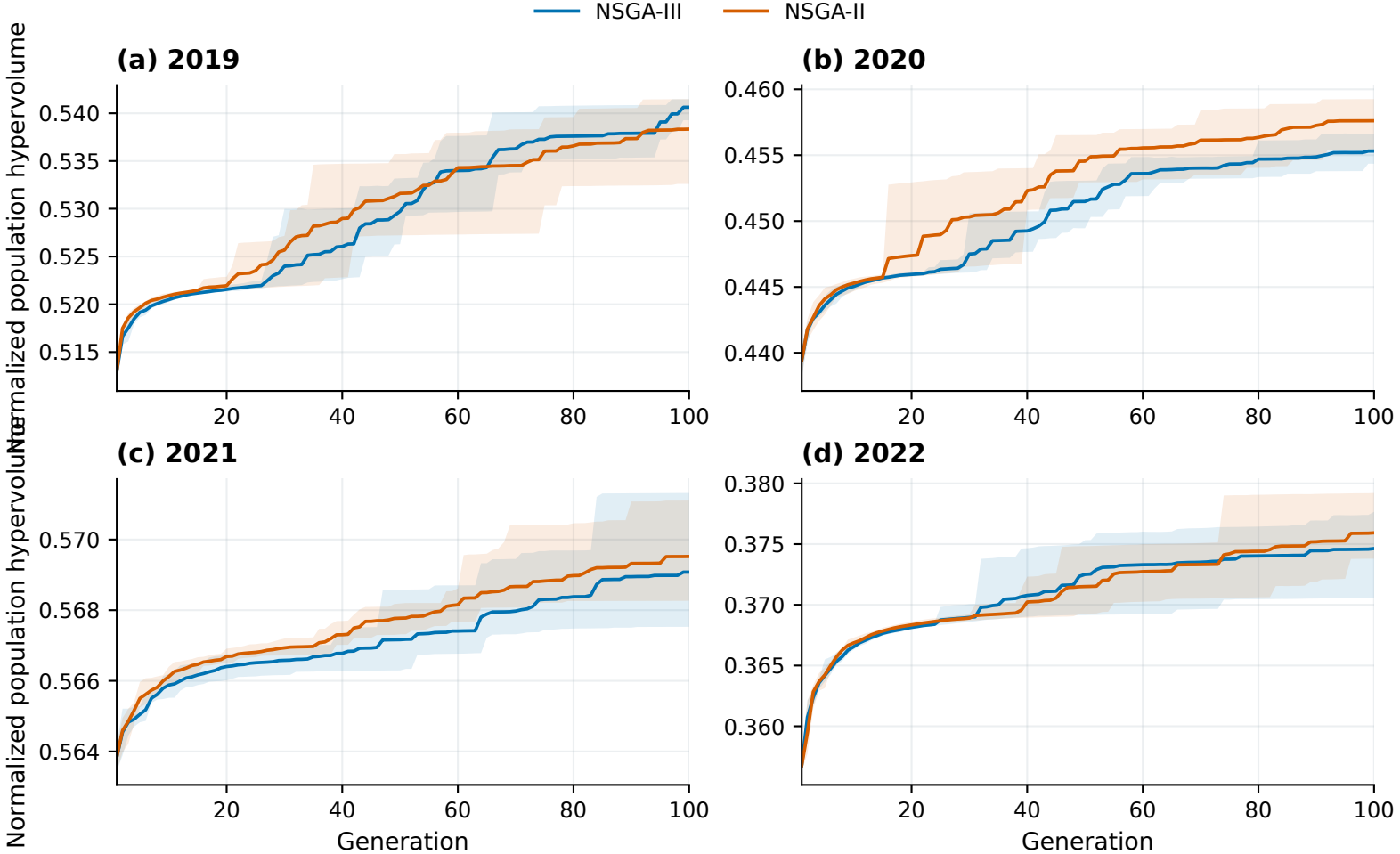

Four-year mean
2019
2020
2021
2022
(a) First-order effect (S_1)
(b) Total effect (S_T)
CCx
CGC
CDC
Senescence
Kcb
WP early
WP late
HI0
0.0
0.1
0.2
0.3
0.4
0.5
Sobol index
Sobol index


NSGA-III front
Evaluated solutions
Original DI model baseline
Selected 95% yield solution
95% observed yield
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Grain yield at 13% moisture (t ha⁻¹)
Actual seasonal irrigation (mm)
(0, 0, 0, 0, 0)
(1, 23, 15, 50, 0)
(68, 16, 59, 35, 0)
(75, 7, 114, 49, 22)
(95, 88, 128, 50, 1)
(92, 87, 80, 113, 73)
(0, 33, 24, 22, 3)
(0, 116, 10, 26, 14)
(112, 102, 3, 100, 13)
(164, 73, 173, 1, 1)
(0, 5, 19, 34, 11)
(0, 101, 2, 23, 13)
(0, 179, 0, 99, 0)
(0, 188, 0, 17, 3)
(49, 141, 12, 140, 6)
(0, 19, 23, 17, 17)
(0, 41, 76, 23, 13)
(0, 174, 34, 97, 2)
(0, 175, 1, 53, 1)
(102, 109, 67, 104, 0)


Lower evapotranspiration
Lower end-of-season storage
Lower other losses
Irrigation reduction components (mm)
0
100
200
300
400
82.0%
92.7%
96.7%
81.1%
2019
2020
2021
2022


Seedling
Jointing
Heading
Grain filling
Maturity
(a) 2019
I = 209.83 mm; Y = 15.725 t ha^{-1}
12 positive-depth events
(b) 2020
I = 306.50 mm; Y = 16.427 t ha^{-1}
13 positive-depth events
(c) 2021
I = 264.50 mm; Y = 15.068 t ha^{-1}
9 positive-depth events
(d) 2022
I = 257.47 mm; Y = 13.390 t ha^{-1}
8 positive-depth events
Individual irrigation depth (mm)
May
Jun
Jul
Aug
Sep


Original DI schedule
Optimized schedule; n = positive-depth events
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Stage irrigation total (mm)
0
50
100
150
200
250
n=2
n=3
n=2
n=3
n=2
n=1
n=3
n=3
n=3
n=2
n=3
n=1
n=2
n=1
n=2
n=1
n=2
n=1
n=2
Seedling
Jointing
Heading
Grain filling
Maturity


NSGA-III
NSGA-II
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Normalized population hypervolume
Normalized population hypervolume
Generation
Generation
0.540
0.535
0.530
0.525
0.520
0.515
0.460
0.455
0.450
0.445
0.440
0.570
0.568
0.566
0.564
0.380
0.375
0.370
0.365
0.360
20
40
60
80
100